Transitivity of derivability in the axiomatic system for first-order logic

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own proposition environment prop:transitivity, which carries the optional title Transitivity (source lines 842-846, the \ollabel on line 843); the proof environment begins two lines below it and is its own proof.

Editorial note. Neither the statement nor the proof contains an \olref or a \cite, so no source pointer and no bibliography entry are needed; every symbol used (\Proves, \Gamma, \Delta, the !A shorthand and !!a{...}) is declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained proposition prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's first-order-logic axiomatic-deduction unit, counted once: derivability in the axiomatic system is transitive in the hypotheses.

Allowed standard prerequisites (editorial).

- Derivation from a set of hypotheses. A derivation of A from Gamma is a finite sequence of formulas ending in A in which every formula is an axiom, an element of Gamma, or follows from earlier lines by a rule of inference.. Both hypotheses are unpacked in this form at source lines 848-856, and the concatenated sequence is then verified to be a derivation in the same sense at lines 856-862.

- Rules of inference and the mixing rule. The axioms and rules of the author's axiomatic system for first-order logic, in particular the rule that licenses a formula whenever a prior line is a conditional whose antecedent is also a prior line.. The proof's key step replaces each use of the hypothesis A in the second derivation by the corresponding step of the first derivation, appealing to the fact that the same rule of inference still justifies the line (source lines 860-862).

Proposition 0.1 (Transitivity)**.** *If* $\Gamma \vdash !A$ *and* $\{!A\} \cup \Delta \vdash !B$*, then* $\Gamma \cup \Delta \vdash !B$*.*

Proof. Suppose $\{!A\} \cup \Delta \vdash !B$. Then there is !!aderivation $!B_1$, ..., $!B_l = !B$ from $\{!A\} \cup \Delta$. Some of the steps in that !!derivation will be correct because of a rule which refers to a prior line $!B_i = !A$. By hypothesis, there is !!aderivation of $!A$ from Γ, i.e., !!aderivation $!A_1$, ..., $!A_k = !A$ where every $!A_i$ is an axiom, !!aelement of Γ, or correct by a rule of inference. Now consider the sequence

$$!A_1, \ldots, !A_k = !A, !B_1, \ldots, !B_l = !B.$$

This is a correct !!derivation of $!B$ from $\Gamma \cup \Delta$ since every $B_i = !A$ is now justified by the same rule which justifies $!A_k = !A$. □